

GenMol UDLM engineering exploration

engineering-v6-gibbs | complete | 2026-09-07T10:09:32.675735+00:00

Engineering pilots only. Superiority has not been established.

Engineering pilots only: no superiority claim, final candidate lock, or final-seed evaluation.

The local MDLM model completed 50,000 updates; each UDLM arm starts from its EMA weights and adds 1,000 updates. Training compute is not matched.

The scheduler and FiLM conditioner were selected on E denoising loss and then shared by R/S/E. This is not independent optimization of each method.

Small samples and configuration selection create sampling uncertainty and selection bias; a higher pilot mean is not evidence of superiority.

Strict decoding uses fix=False. Released-compatible decoding uses SAFE repair and largest-component selection; report both because repairs can hide malformed outputs.

Validity and quality divide by requested samples. Uniqueness divides by valid samples. Quality counts unique molecules with QED ≥ 0.6 and SA ≤ 4 . Diversity is the released PyTDC fingerprint metric over unique valid molecules.

Means and sample standard deviations average seeds equally within one configuration. Diversity is not pooled across seeds. Failed, pending, and invalid runs are excluded from means and remain disclosed.

Paper seeds are undisclosed and hardware differs. Local timings are descriptive; no speed claim relative to the paper is made.

Prospective sampling design

```
{"arms": ["R", "S", "E"], "claim_boundary": "Learned/tempered LOO conditionals are approximate; exact Gibbs stationarity is established only for the toy true-conditional tests.", "corrector_coordinate_selection": "one uniformly selected editable coordinate per row, fresh prediction at the post-predictor state and time", "frozen_before_v6_generation": true, "gibbs_treatment": {"corrector_updates": 64, "nfe": 128, "predictor_transitions": 64}, "partial_v5_observed_before_design": "E and S temperature outcomes through 0.85; no final evaluation seeds. Temperature 0.5 is fixed across arms to favor syntactic fidelity, not selected as the global quality winner.", "predictor_control": {"corrector_updates": 0, "nfe": 128, "predictor_transitions": 128}, "temperature": 0.5, "top_p": 1.0}
```

Exploratory parameter selection, not a confirmatory benchmark.

Only 1000 UDLM continuation updates: matched method training superiority is not isolated.

Historical baseline has 1000 samples per seed versus 64 here; diversity and uniqueness vary with sample count.

Strict molecular validity is reported separately from repaired validity.

The corrector changes transition allocation at fixed total NFE, not training.

Temperature 0.5 is a common engineering choice informed by partial v5 syntax diagnostics.

Corrector predictions can depend on their own noisy token; exact stationarity is not asserted for the learned model.

Reference means (context only)

Reference	validity	uniqueness	quality	diversity
local_mdln_50000	1.000000	0.998667	0.858000	0.823021
paper_genmol_v1_table1	1.000000	0.997000	0.846000	0.818000

Sampling evaluation budget

Counts are per molecule. Each predictor transition and each fresh Gibbs correction consumes one backbone evaluation. Predictor-only counts equal recorded NFE; missing runs have no observed budget.

Config	Seed	Total NFE	Predictors	Correctors
e_t050_predictor	1300	128	128	0
e_t050_predictor	1301	128	128	0
e_t050_gibbs	1300	128	64	64
e_t050_gibbs	1301	128	64	64
s_t050_predictor	1300	128	128	0
s_t050_predictor	1301	128	128	0
s_t050_gibbs	1300	128	64	64
s_t050_gibbs	1301	128	64	64
r_t050_predictor	1300	128	128	0
r_t050_predictor	1301	128	128	0
r_t050_gibbs	1300	128	64	64
r_t050_gibbs	1301	128	64	64

Per-configuration mean, equal seed weights

released_comparable

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.9688 +/- 0.0000	0.9919 +/- 0.0114	0.4688 +/- 0.0884	0.8959 +/- 0.0102
e_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9841 +/- 0.0004	0.5156 +/- 0.0663	0.8886 +/- 0.0033

Config	Seeds complete	validity	uniqueness	quality	diversity
s_t050_predictor	2/2	0.9844 +/- 0.0221	0.9919 +/- 0.0114	0.5469 +/- 0.0442	0.8822 +/- 0.0064
s_t050_gibbs	2/2	0.9844 +/- 0.0221	0.9761 +/- 0.0118	0.5703 +/- 0.0110	0.8921 +/- 0.0052
r_t050_predictor	2/2	1.0000 +/- 0.0000	0.9844 +/- 0.0221	0.5391 +/- 0.0552	0.8850 +/- 0.0070
r_t050_gibbs	2/2	0.9766 +/- 0.0110	1.0000 +/- 0.0000	0.4766 +/- 0.0331	0.8932 +/- 0.0124

strict

Config	Seeds complete	validity	uniqueness	quality	diversity
e_t050_predictor	2/2	0.5391 +/- 0.0552	1.0000 +/- 0.0000	0.3125 +/- 0.0221	0.8874 +/- 0.0056
e_t050_gibbs	2/2	0.5547 +/- 0.0994	1.0000 +/- 0.0000	0.2969 +/- 0.0000	0.8744 +/- 0.0029
s_t050_predictor	2/2	0.6172 +/- 0.0110	1.0000 +/- 0.0000	0.3359 +/- 0.0110	0.8659 +/- 0.0067
s_t050_gibbs	2/2	0.5547 +/- 0.0773	1.0000 +/- 0.0000	0.3672 +/- 0.0110	0.8776 +/- 0.0045
r_t050_predictor	2/2	0.6562 +/- 0.0884	1.0000 +/- 0.0000	0.3672 +/- 0.0331	0.8743 +/- 0.0009
r_t050_gibbs	2/2	0.6172 +/- 0.0773	1.0000 +/- 0.0000	0.3203 +/- 0.0552	0.8769 +/- 0.0080

Every scheduled seed

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
e_t050_predictor	1300	completed	64	64	0.5312	0.3281	9.11
e_t050_predictor	1301	completed	64	64	0.4062	0.2969	7.44
e_t050_gibbs	1300	completed	64	64	0.5625	0.2969	9.46
e_t050_gibbs	1301	completed	64	64	0.4688	0.2969	8.86
s_t050_predictor	1300	completed	64	64	0.5156	0.3281	9.25
s_t050_predictor	1301	completed	64	64	0.5781	0.3438	8.94
s_t050_gibbs	1300	completed	64	64	0.5625	0.3594	9.59
s_t050_gibbs	1301	completed	64	64	0.5781	0.3750	8.90

Config	Seed	Status	Requests	Raw rows	Repair Q	Strict Q	Generation s
r_t050_predictor	1300	completed	64	64	0.5781	0.3906	9.24
r_t050_predictor	1301	completed	64	64	0.5000	0.3438	8.90
r_t050_gibbs	1300	completed	64	64	0.5000	0.3594	9.31
r_t050_gibbs	1301	completed	64	64	0.4531	0.2812	8.81

Preserved v4 diagnostic failure

V4 remains campaign_incomplete; its failed D attempt is never reclassified or reused for ranking.

completion validation failed: Seed 1100 has benchmark artifacts that do not match the requested run or fail integrity checks: - config.effective={'device': 'cuda:0', 'diffusion_type': 'udlm', 'exclude_special_tokens': False, 'inference_eps': 1e-05, 'min_add_len': 40, 'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'num_steps': 128, 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'prior_variant': 'empirical_frequency', 'randomness': 0.0, 'raw_loo_top_p': 1.0, 'softmax_temp': 1.0}; expected {'model_path': '/home/aidar.alimbayev/Documents/genmolv2/run_sources/udlm_genmol_worktree/output/udlm/scaleup-w1-e-dcb271453411/checkpoints/1000.ckpt', 'num_samples': 32, 'diffusion_type': 'udlm', 'softmax_temp': 1.0, 'randomness': 0.0, 'min_add_len': 40, 'num_steps': 128, 'inference_eps': 1e-05, 'exclude_special_tokens': False, 'prior_variant': 'empirical_frequency', 'prior_metadata_sha256': 'f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612', 'device': 'cuda:0'} - config.effective_sha256='278d4476e98bc0b5905e16c2158dd75a3a5a85250e4a48726640afc0e5529608'; expected '3ee30748fa7d7fcc675fde0fe1e8ff15f23fdc3374ba76eed03466ba306ec02f' Refusing to skip or relaunch into the existing directory. Inspect the artifacts or choose a fresh output root.

Configuration and artifact provenance

Protocol SHA-256: b50d97dcdd10f35c85aba61c9b9b948081ae9726809abb25b6c9e0a8834a92e9

v6-e-t050-predictor / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index":

"5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1300/raw_samples.csv", "sha256": "e65e1f806e958601528d3c48b31e727f004318f427956ba7dd77ceece5d3b5d2", "size_bytes": 20407}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-predictor / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 343481, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 872}], "index": 6, "memory_total_mib": 49140, "memory_used_mib": 895, "name": "NVIDIA RTX A6000", "utilization_percent": 0, "uuid": "GPU-d7e40ce5-9260-a208-16a0-3372e3f204de"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "6"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-predictor/seed_1301/raw_samples.csv", "sha256": "30f4f1668bd4716728fe9fed2adbe5efc397a709c75716a17f57c8f0c57b167", "size_bytes": 20096}, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1300: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e3ce", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.rokettishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e3ce"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "7912f9c622d82870eddb211c1688a73fb100b0cefd49e8a614a8622ae88db94b", "size_bytes": 19526}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-e-t050-gibbs / seed 1301: {"checkpoint": "dce870e8d63453f73c428b9115f33556f67a21d45788d3112c2777ea82623005", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "raw_loo_top_p": 1.0,

"softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "f738b8b17de5c4704058018bbddacd7fed779c85248e33d199b68a648151e612", "prior_variant": "empirical_frequency", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-e-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "64c88ce12d8ac7a022d6b1715cbef0ffa2ec0434145bae6f9995895d94eb492f", "size_bytes": 20839, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e6ce", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fd6e6ce"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1300/raw_samples.csv", "sha256": "cfe608dd00991b6b95db83694542150cc55feeabfa262953d9a312f317ea3017", "size_bytes": 20527, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-predictor / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes":

{["pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880]}, "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-predictor/seed_1301/raw_samples.csv", "sha256": "df3e72494039fd6a19e51489cfdea60f69d74bea55deee8f94c44b6d19c20ff0", "size_bytes": 20731, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1300: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2", "raw": {"relative_path": "output/udlm/engineering_v6/v6-s-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "3d389848d51f373a2f911fb06c34048f23276cef913b5acb201495d40cfdb0d1", "size_bytes": 20711, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-s-t050-gibbs / seed 1301: {"checkpoint": "100f467b94766f2c87cc734398c8590bd14a1c8e0722ce31dbca7b446d986e72", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": "a5bbbe437cb732c516e3f2ad461b66de7f55e2e160e02c724f5e778cffb46fb7", "prior_variant": "schedule_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 6, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name":

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v6-r-t050-predictor / seed 1300: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "gpu": {"cuda_visible_devices": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 275418, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 5, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "5", "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-predictor/seed_1300/raw_samples.csv", "sha256": "72525be9b9e5ccfafa3c4af4a1310f56328499f291dfe5290d57789777f52d2", "size_bytes": 20472, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}}

v6-r-t050-predictor / seed 1301: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "generation_protocol": {"diffusion_type": "udlm", "exclude_special_tokens": false, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per reverse step", "num_steps": 128, "num_steps_source": "explicit UDLM reverse-transition count", "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than", "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216, "physical_index": "2", "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-predictor/seed_1301/raw_samples.csv", "sha256": "af6ecd7bbb3baad91217fe15a63e7fc88c2a06d1713b0cfe2ca71035ff6a22c2", "size_bytes": 21730, "sampling_budget": {"corrector_steps_per_molecule": 0, "gibbs_corrector": false, "nfe": 128, "predictor_transitions_per_molecule": 128}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}}

v6-r-t050-gibbs / seed 1300: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation

per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 340770, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 880}], "index": 2, "memory_total_mib": 49140, "memory_used_mib": 903, "name": "NVIDIA RTX A6000", "utilization_percent": 8, "uuid": "GPU-997e881a-bd08-aa54-45f3-9b3c6fdf6ece"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "2"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1300/raw_samples.csv", "sha256": "82a3ed576ca875885098408c9279a0b75d1d33d4590b7d82bcb0ea23d8f32823", "size_bytes": 20943}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}

v6-r-t050-gibbs / seed 1301: {"checkpoint": "d0310d2e2402043ab9ab6a99268263581a35d60fb2262ddd988ea38cd6f6395c", "config": {"diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "min_add_len": 40, "num_steps": 128, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "raw_loo_top_p": 1.0, "softmax_temp": 0.5}, "corrector_source_sha256": "d912a0e063e8d2a86170e2b703f428cdcb628fa173f9c129e9359e19824a094f", "generation_protocol": {"corrector_steps_per_molecule": 64, "diffusion_type": "udlm", "exclude_special_tokens": false, "gibbs_corrector": true, "inference_eps": 1e-05, "inference_weights": {"ema": {"decay": 0.9999, "num_updates": 1000, "shadow_parameter_count": 230}, "ema_applied": true, "source": "ema"}, "model_use_bracket_safe": false, "nfe": 128, "nfe_definition": "one full backbone forward evaluation per predictor transition and per fresh Gibbs corrector", "num_steps": 128, "num_steps_source": "explicit UDLM total predictor-plus-corrector NFE budget", "predictor_transitions_per_molecule": 64, "prior_metadata_sha256": null, "prior_variant": "release_uniform", "randomness": 0.0, "randomness_used_by_sampler": false, "raw_loo_top_p": 1.0, "released_largest_component": "maximum SMILES string length", "released_safe_fix": true, "single_generation_batch": true, "strict_safe_fix": false, "temperature": 0.5}, "gpu": {"cuda_visible_devices": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959", "final_probe": {"compute_mode": "Default", "compute_processes": [{"pid": 227850, "process_name": "/home/meir.roketlishvili/miniconda3/envs/SGPO_repro/bin/python", "used_memory_mib": 1284}], "index": 3, "memory_total_mib": 49140, "memory_used_mib": 1307, "name": "NVIDIA RTX A6000", "utilization_percent": 7, "uuid": "GPU-2cd1aa5b-616e-e6d3-b54d-49241cc8f959"}, "launch_policy": {"active_compute_processes_allowed": true, "inventory_scope": "all_nvidia_gpus", "max_utilization_percent": 10, "min_free_memory_mib": 30000, "requested_gpu_count": 2, "selection_method": "dynamic_idle_discovery", "utilization_comparison": "strictly_less_than"}, "logical_device": {"compute_capability": [8, 6], "logical_index": 0, "name": "NVIDIA RTX A6000", "total_memory_bytes": 50897289216}, "physical_index": "3"}, "raw": {"relative_path": "output/udlm/engineering_v6/v6-r-t050-gibbs/seed_1301/raw_samples.csv", "sha256": "a7f427c0741144239f64a594a7cca57bdbb98d8297cf62a2dbfb7065105dc678", "size_bytes": 20300}, "sampling_budget": {"corrector_steps_per_molecule": 64, "gibbs_corrector": true, "nfe": 128, "predictor_transitions_per_molecule": 64}, "source": "84e909d983c0c413a9bce71357deba24df717b85"}